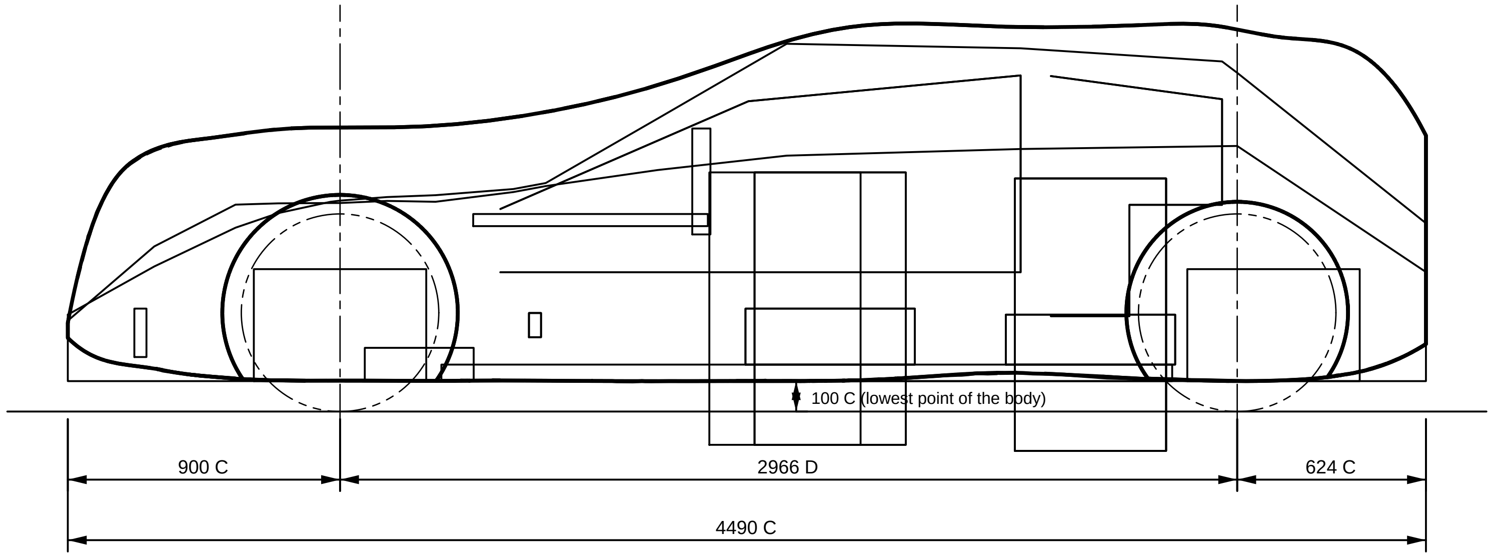
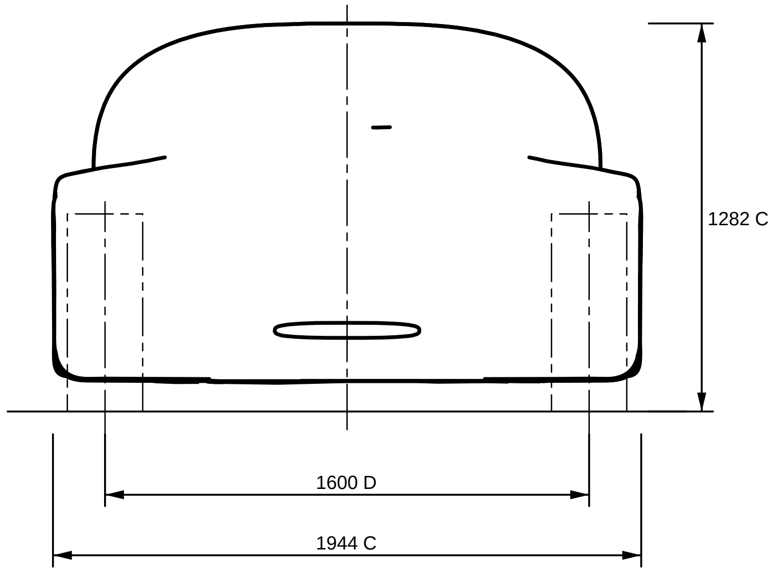
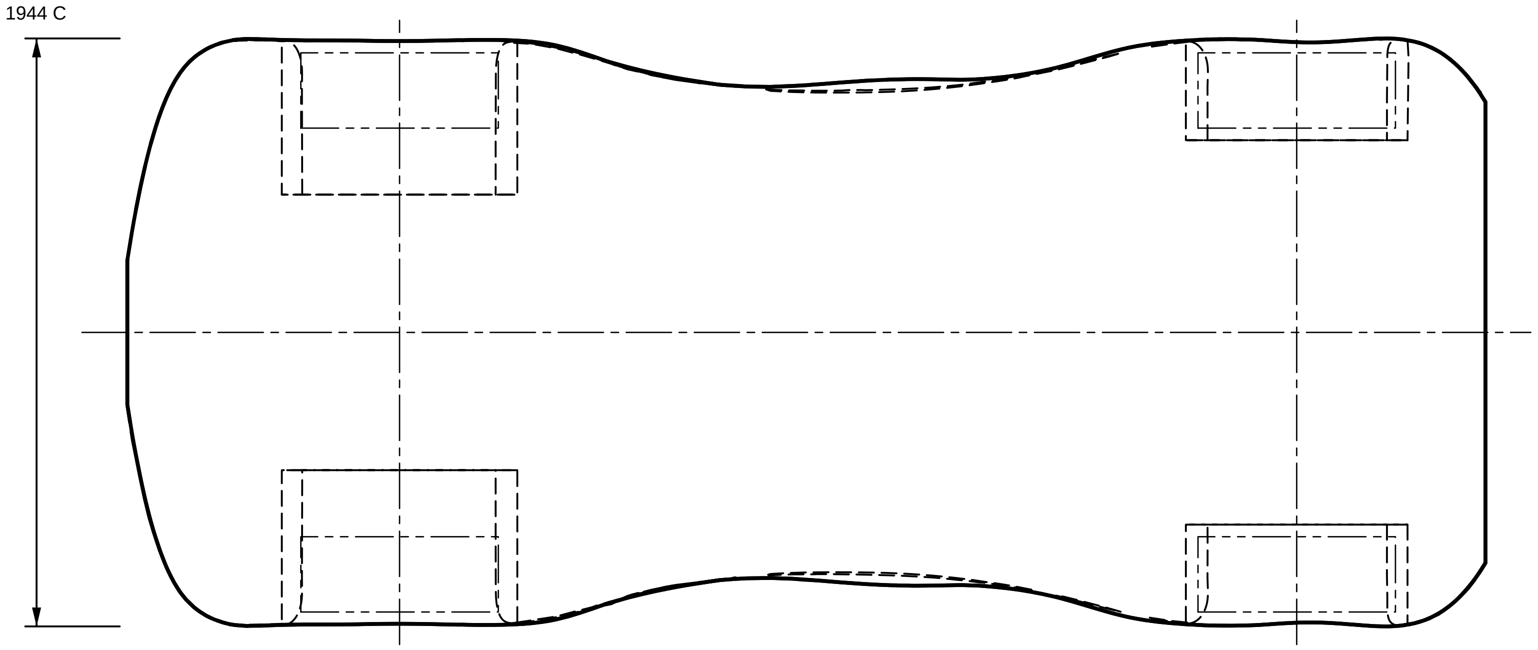
SIDE ELEVATION (left side)



FRONT VIEW



PLAN (from above; the car's right side at the top)



NOTES

Dimensions in mm. C = computed from the CAD body (kernel receipt), D = layout design choice or derivation, S = sourced (cited). Values and bases: sheet 2.

Hidden-line projection of the B-rep body (OpenCascade polygonal HLR); edges within the tessellation deflection (2 mm). Tyres and axles: phantom / centre lines.

Cutaway lines (thin) are the assembly projection: occupants, openings, controls and powertrain envelopes.

Revision = hash of the model this sheet was generated from. Any change to the body, layout or BOM gives a new revision; a sheet whose revision differs from the model's current one is superseded.

SCREENING DRAWING: generated from a software model; not checked, not approved, not for manufacture.

All masses are mass.part results (state: sourced / estimated / computed / placeholder); occupants (payload) are not parts and are not lis...

GENERAL ARRANGEMENT - VEH (ConKay library car)	
Drawing no. CK-GA-VEH Sheet 1 of 3	Revision
Scale 1:25 (A3) Units mm First-angle projection	**R-56D8A632**
Model: BODY_SHELL + VEH layout + BOM (56 items)	model hash 56d8a632fd987325b5afadf6929c7d0c
SCREENING - NOT FOR MANUFACTURE Generated by ConKay drawing.ga 1.0.0 from cad.body (OCCT 7.7.2.0); revision = model hash	f5c6191ef983a1e7a24db19df95f6c84

BILL OF MATERIALS (component library + designed parts)

Item	Qty	Nodes	Description	Library id / material	Unit mass kg	Mass state	Source
1	1	BODY_SHELL	BODY_SHELL (designed part)	material: cfrp-hexply-8552-agp193pw	120.52	computed	cfrp-hexply-8552-agp193pw (density: Hexcel HexPly 8552 product...
2	1	BRAKES_FRONT	Wilwood AERO6 Big Brake front brake kit 140-13886, black calipers, 14...	brakes.wilwood.aero6.140-13886	23.86	sourced	www.wilwood.com/BrakeKits/BrakeKitsProdFront
3	1	BRAKES_REAR	Wilwood AERO4-MC4 Big Brake rear parking brake kit 140-13888, black c...	brakes.wilwood.aero4-mc4.140-13888	20.32	sourced	www.wilwood.com/BrakeKits/BrakeKitsProdRear
4	1	CONTROL_PACK	Ford Performance Gen 4X Coyote control pack, manual transmission M-60...	wiring.ford.control-pack.m-6017-m50hm	14.06	estimated	performanceparts.ford.com/part/M-6017-M50HM
5	1	DIFFERENTIAL	Ford Performance Mustang Super 8.8-inch IRS loaded differential housi...	differential.ford.super-8.8-irs.m-4001-88355b	71.89	estimated	www.mustang6g.com/forums/threads/the-what-things-weigh-th
6	1	ENGINE	Ford Performance 5.0L Gen 4X Mustang crate engine M-6007-M50H (10R80 ...	engine.ford.coyote-gen4x.m-6007-m50h	201.85	sourced	www.ford.com/product/50l-gen-4x-mustang-crate-engine
7	1	EXHAUST	Ford (OEM) / Bassani Exhaust after the crate engine's right-hand mani...	exhaust.s550-gt.oem-lh-manifold-midpipe.bassani-xpipe-catback	36.02	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
8	1	FUEL_CELL	ATL (Aero Tec Laboratories) Saver Cell, 60 L (15 US gal) SA-AA-070 (l...	fuel.atl.saver-cell.sa-aa-070	8.10	sourced	atlinside.com/Content/Downloads/ATL-Saver-Cell-Data-S
9	1	GLAZING	GLAZING (designed part)	material: glass-soda-lime	22.00	computed	glass-soda-lime (density: typical annealed float glass; streng...
10	1	RADIATOR	Cold Case Radiators Coyote swap aluminium radiator kit with fans and ...	cooling.coldcase.lmm570-5k	16.78	estimated	wheelfront.com/shop/product/cold-case-radiators-79-93-
11	2	SEAT_1, SEAT_2	RECARO Sportster GT reclining sport seat, metal seat base with reinfo...	seat.recaro.sportster-gt	15.00	sourced	www.recaro-automotive.com/fileadmin/01-products/documents/hotshee
12	2	SEAT_3, SEAT_4	Ford (OEM) S550 Mustang rear seat 2015 Mustang OEM rear seat backs + ...	seat.ford.s550-rear.oem	9.48	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
13	1	STEERING_RACK	Ford (OEM) S550 Mustang electric power-assisted steering (EPAS) rack ...	steering.ford.s550-epas	18.35	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
14	1	SUSPENSION_FRONT	Ford (OEM) S550 Mustang front suspension 2015 Mustang GT (non-Perform...	suspension.ford.s550-front.oem	78.06	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
15	1	SUSPENSION_REAR	Ford (OEM) S550 Mustang independent rear suspension 2015 Mustang GT O...	suspension.ford.s550-irs-rear.oem	61.83	sourced	www.mustang6g.com/forums/threads/the-what-things-weigh-th
16	4	TIRE_FL, TIRE_FR, TIRE_RL, TIRE_RR	Michelin Pilot Sport 4S 245/40ZR18 (97Y) XL, MSPN 37666	tyre.michelin.pilot-sport-4s.245-40zr18-97y-xl	11.52	sourced	wheelwiz.ca/products/m-37666
17	1	TRANSMISSION	TREMEC TKX 5-speed RWD manual TCET18085 (Ford, 26-spline input, 3.27 ...	transmission.tremec.tkx.tcet18085	50.00	estimated	tremec.com/wp-content/uploads/2023/05/TKX-Service-
18	1	TUB_A_PILLAR_L	A-pillar L (designed part)	material: aluminum-6061-t6	3.22	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
19	1	TUB_A_PILLAR_R	A-pillar R (designed part)	material: aluminum-6061-t6	3.22	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
20	1	TUB_B_POST_L	B-post L (designed part)	material: aluminum-6061-t6	3.15	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
21	1	TUB_B_POST_R	B-post R (designed part)	material: aluminum-6061-t6	3.15	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
22	1	TUB_C_POST_L	C-post L (designed part)	material: aluminum-6061-t6	2.19	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
23	1	TUB_C_POST_R	C-post R (designed part)	material: aluminum-6061-t6	2.19	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
24	1	TUB_FIREWALL_L	Firewall sheet L (designed part)	material: aluminum-6061-t6	1.48	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
25	1	TUB_FIREWALL_R	Firewall sheet R (designed part)	material: aluminum-6061-t6	1.48	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
26	1	TUB_FLOOR_L	Floor sheet L (designed part)	material: aluminum-6061-t6	5.40	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
27	1	TUB_FLOOR_R	Floor sheet R (designed part)	material: aluminum-6061-t6	5.40	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
28	1	TUB_FW_CPOST	Firewall centre post (designed part)	material: aluminum-6061-t6	0.55	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
29	1	TUB_FW_LOWER	Firewall lower cross (designed part)	material: aluminum-6061-t6	17.45	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
30	1	TUB_FW_POST_L	Firewall post L (designed part)	material: aluminum-6061-t6	0.73	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
31	1	TUB_FW_POST_R	Firewall post R (designed part)	material: aluminum-6061-t6	0.73	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
32	1	TUB_HEEL_CROSS	Heel cross (designed part)	material: aluminum-6061-t6	3.87	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
33	1	TUB_KICK_L	Sill kick L (designed part)	material: aluminum-6061-t6	4.80	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
34	1	TUB_KICK_R	Sill kick R (designed part)	material: aluminum-6061-t6	4.80	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
35	1	TUB_QTR_L	Rear quarter box L (designed part)	material: aluminum-6061-t6	7.97	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
36	1	TUB_QTR_R	Rear quarter box R (designed part)	material: aluminum-6061-t6	7.97	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
37	1	TUB_RAIL_FL	Front rail L (designed part)	material: aluminum-6061-t6	4.60	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
38	1	TUB_RAIL_FR	Front rail R (designed part)	material: aluminum-6061-t6	4.60	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
39	1	TUB_REAR_BULKHEAD	Rear bulkhead (designed part)	material: aluminum-6061-t6	5.52	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
40	1	TUB_REAR_CROSS	Rear end cross (designed part)	material: aluminum-6061-t6	4.84	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
41	1	TUB_REAR_FLOOR_L	Rear floor sheet L (designed part)	material: aluminum-6061-t6	1.66	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
42	1	TUB_REAR_FLOOR_R	Rear floor sheet R (designed part)	material: aluminum-6061-t6	1.66	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
43	1	TUB_ROOF_BOW	Roof bow (B-hoop top) (designed part)	material: aluminum-6061-t6	3.56	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
44	1	TUB_ROOF_LINER	Roof liner (bonded to the skin) (designed part)	material: aluminum-6061-t6	4.00	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
45	1	TUB_ROOF_RAIL_L	Roof rail L (designed part)	material: aluminum-6061-t6	3.27	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
46	1	TUB_ROOF_RAIL_R	Roof rail R (designed part)	material: aluminum-6061-t6	3.27	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
47	1	TUB_ROOF_RAIL_R_L	Rear roof rail L (designed part)	material: aluminum-6061-t6	2.32	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
48	1	TUB_ROOF_RAIL_R_R	Rear roof rail R (designed part)	material: aluminum-6061-t6	2.32	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
49	1	TUB_SCUTTLE	Scuttle cross (designed part)	material: aluminum-6061-t6	4.20	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
50	1	TUB_SEAT_CROSS	Seat cross (designed part)	material: aluminum-6061-t6	3.01	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
51	1	TUB_SILL_L	Sill L (designed part)	material: aluminum-6061-t6	11.57	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
52	1	TUB_SILL_R	Sill R (designed part)	material: aluminum-6061-t6	11.57	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
53	1	TUB_TUNNEL_F	Tunnel, gearbox cell (designed part)	material: aluminum-6061-t6	4.41	computed	aluminum-6061-t6 (density: engineering material library (lib/a...
54	1	TUB_TUNNEL_R	Tunnel, driveshaft cell (designed part)	material: aluminum-6061-t6	5.62	computed	aluminum-6061-t6 (density: engineering material library (lib/a...

GENERAL ARRANGEMENT - VEH (ConKay library car)

Drawing no. CK-GA-VEH Sheet 2 of 3

Scale 1:25 (A3) Units mm First-angle projection

Model: BODY_SHELL + VEH layout + BOM (56 items)

SCREENING - NOT FOR MANUFACTURE

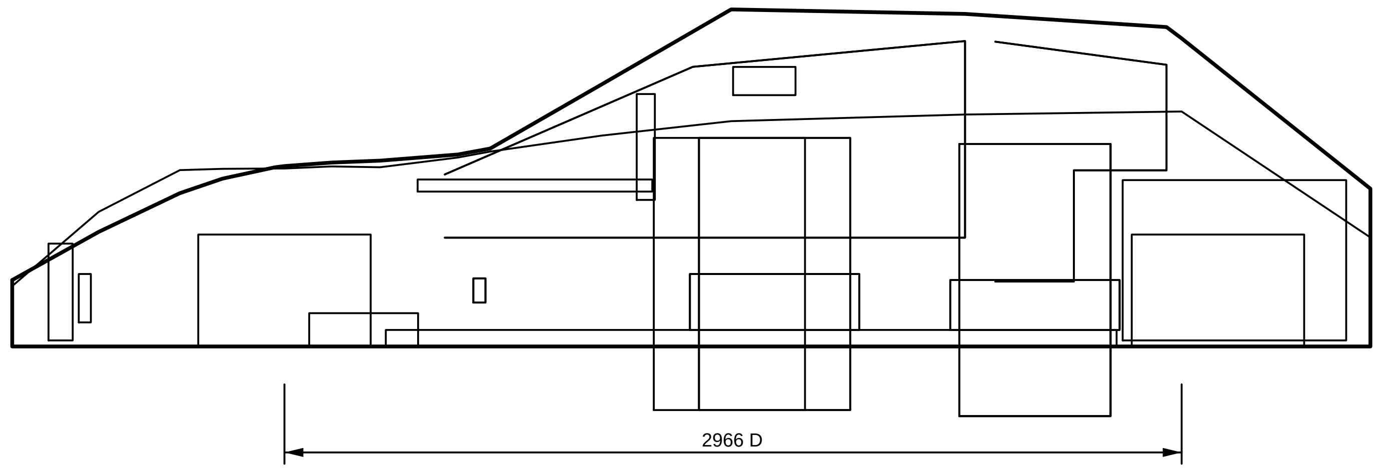
Generated by ConKay drawing.ga 1.0.0 from cad.body (OCCT 7.7.2.0); revision = model hash

Revision

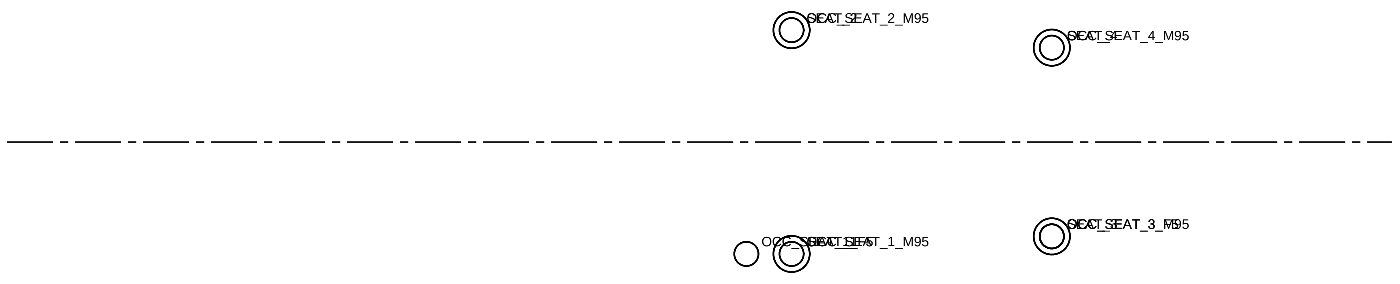
R-56D8A632

model hash 56d8a632fd987325b5afadf6929c7d0c
f5c6191ef983a1e7a24db19df95f6c84

SIDE CUTAWAY (assembly projection, left side)



PLAN - SEAT H-POINTS (assembly)



PACKAGING DIMENSIONS AND FIT

H30 185 mm assumed SAE J1100 (2001)
L53 820 mm calculated SAE J1100 (2001)
L40 25 deg assumed SAE J1100 (2001)
H61-roof 928 mm calculated SAE J1100 (2001)
BOF 203 mm sourced SAE J1100 (2001)
W9 350 mm assumed SAE J1100 (2001)
H17 605 mm assumed SAE J1100 (2001)
eyellipse-X 206.4 mm sourced SAE J941 (2010) Table 2
eyellipse-centroid-z 978 mm sourced SAE J941 (2010) Eq. 4
belt 845 mm calculated ANSUR II acromion, sitting

FIT CHECKS

PASS head-M95 M95 head top at 1139 mm (x 2993 mm), roof at that station 1219 mm, inner gap 76 mm
PASS head-F5 F5 head top at 997 mm (x 2777 mm), roof at that station 1238 mm, inner gap 237 mm
PASS head-M95-rear rear M95 head top at 1119 mm (x 3854 mm), roof at that station 1174 mm, inner gap 51 mm
PASS heel-floor driver heel z 155 mm, floor 155 mm
PASS rear-leg M95 rear occupant reaches the floor in the vertical-shin posture
PASS four-seats 4 seat positions placed
PASS powertrain-cabin MOTOR_FRONT and MOTOR_REAR clear the firewall at 1462 mm
PASS powertrain-layout This design is the light four-seat futuristic car. That shape is a floor battery and compact motors, not a tall front en...
PASS wheel-footwell front tyre swept +/-39.8 deg vs the driver foot, gap 278 mm (separating-axis lower bound)
PASS ingress skin door aperture (roof rail) 1246 mm, M95 head top 1139 mm, gap 107 mm. Tub outline top is 1110 mm; the skin cut is the roof rai...
PASS containment every non-exterior bounding box is inside the body envelope
PASS b-pillar B-pillar station 3250 mm, front H 2620 mm, rear H 3481 mm
WARN cooling-power Cooling core is an assumed EV heat exchanger, not the library gasoline radiator and not a heat-rejection solve. Installed mo...
WARN spare no spare tire is specified, so none is placed
INFO design-quality 33 styling controls. dash-to-axle 23% of the wheelbase, hood 12 mm vs the arch top, rake 60.0 deg, greenhouse/body 0.50, tu...

Projected from the assembly. Envelopes are not kernel solids.

Packaging and occupant layout

Drawing no. CK-GA-VEH Sheet 3 of 3	Revision
Scale 1:25 (A3) Units mm First-angle projection	**R-56D8A632**
Model: VEH assembly	model hash 56d8a632fd987325b5afadf6929c7d0c
SCREENING - NOT FOR MANUFACTURE Projected from assembly.vehicle; revision = model hash	f5c6191ef983a1e7a24db19df95f6c84